

# Using Scratch

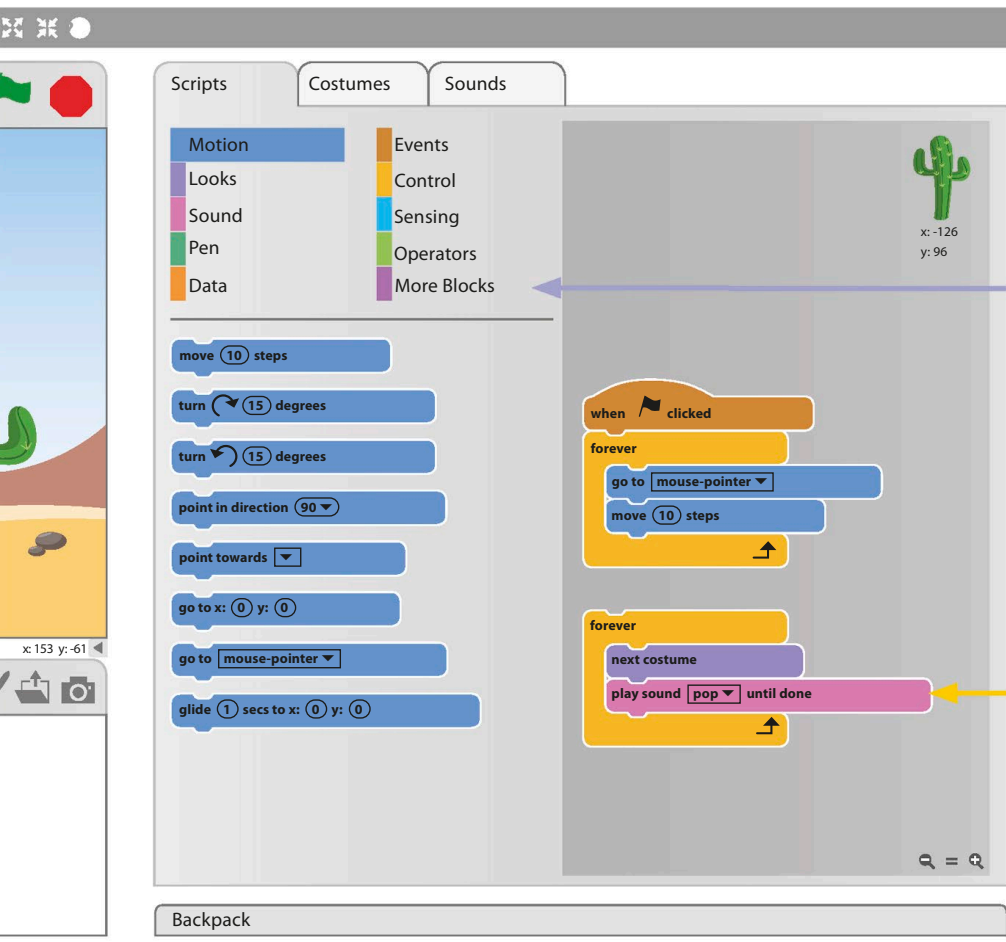
Scratch is an excellent language for new programmers of all ages. It allows them to grasp the basic concepts of programming without the frustration of errors caused by mistyping commands. Scratch's ethos of "remixing" other users' shared code to create new projects also encourages exploration.

| Advantages                                                                    | Disadvantages                                                                |
|-------------------------------------------------------------------------------|------------------------------------------------------------------------------|
| Gives immediate and appealing feedback in the form of animations and sounds   | Only supports a limited range of computational concepts                      |
| Ability to see and modify other users' shared code is a great aid to learning | Not suitable for more advanced programming, as it doesn't include functions. |
| Doesn't require typing skills or memorization of commands                     | Restricts users' ability to write programs that integrate with other systems |

## REAL WORLD

### Controlling code

Scratch allows users to control their programs by moving their body, either by using their computer's webcam or with devices like the Kinect games controller or the Leap motion sensor. This opens up possibilities, such as creating a game that users play without touching a keyboard or another type of controller. They can even create a new musical instrument that's played by dancing.



### Extending Scratch

There are several extensions to Scratch that allow it to be used beyond its original scope. These include blocks to control motors, LEDs, and more from the Raspberry Pi and ScratchX, a suite of experimental blocks that lets users control robots and other devices.

### Blocks

The instruction blocks fit together to make a script that controls a sprite. Blocks doing similar tasks are all one colour; for example, "Sound" blocks are pink, and are labelled with what it does.